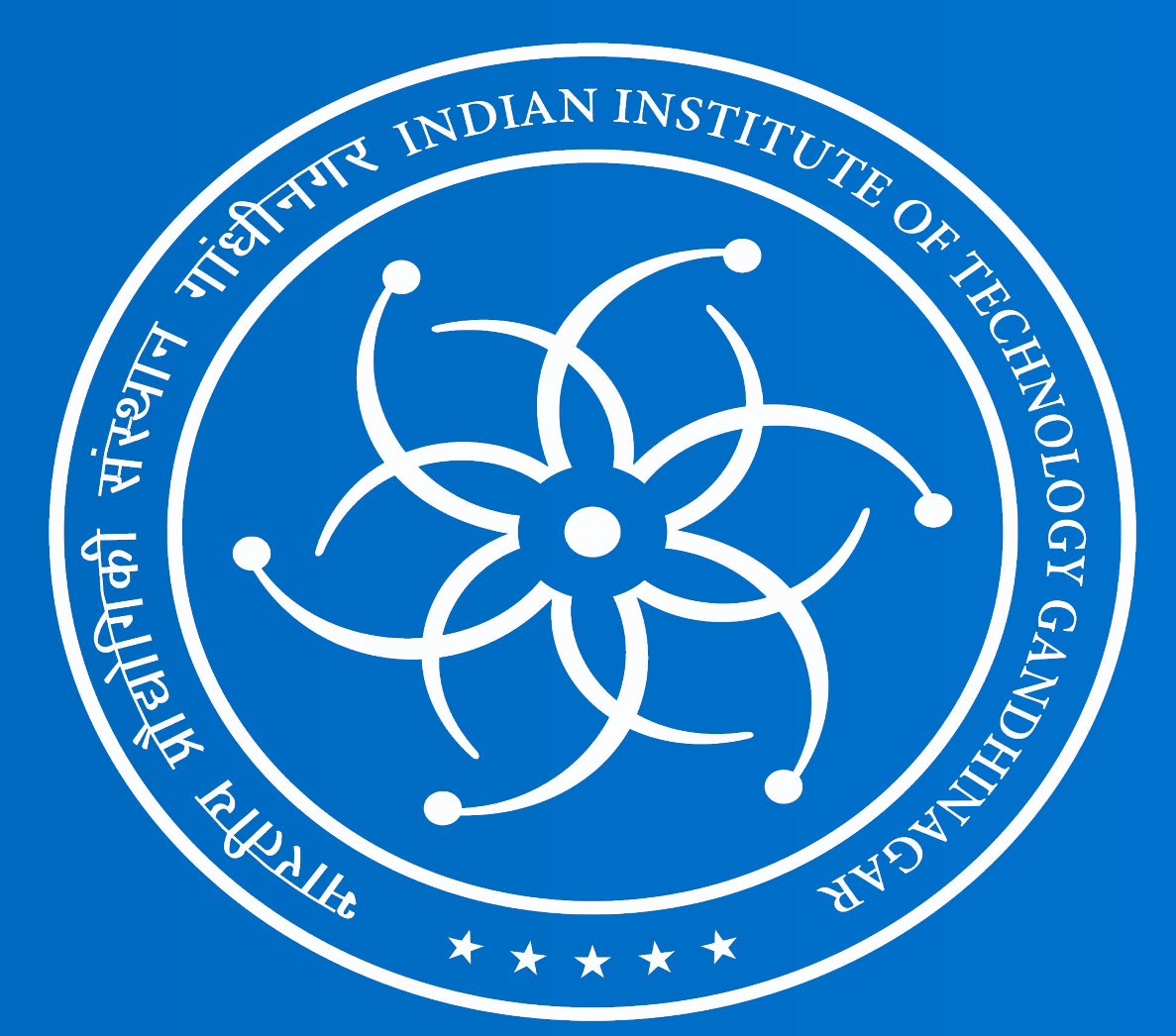


Integrated CFD-ML Framework for Simulation and Dynamic Mode Decomposition for Flow Over Airfoils



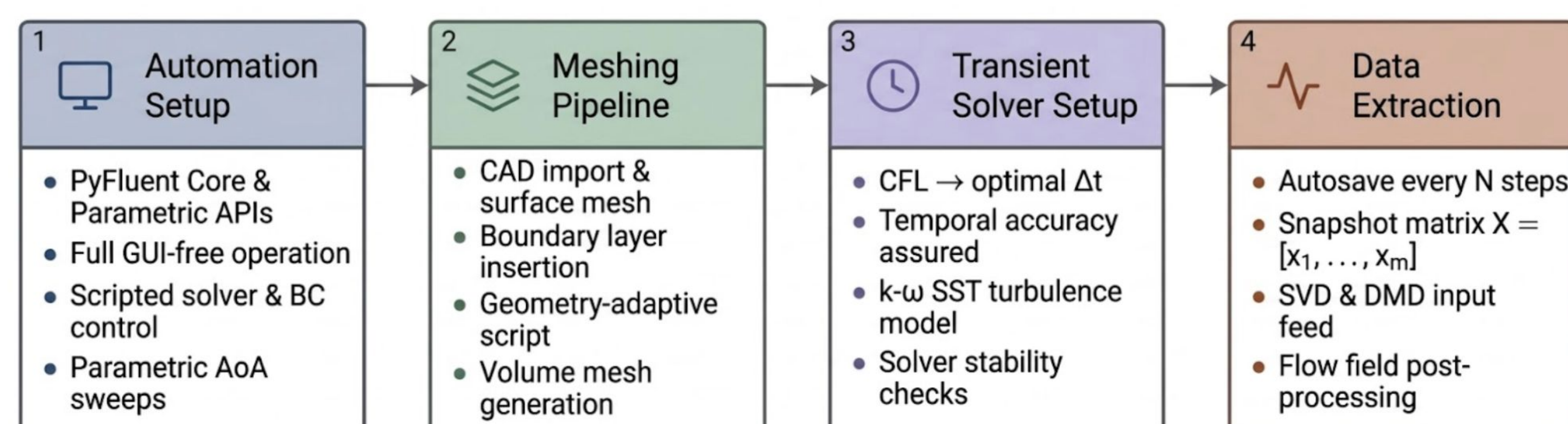
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ME 299: Project Course
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Objective

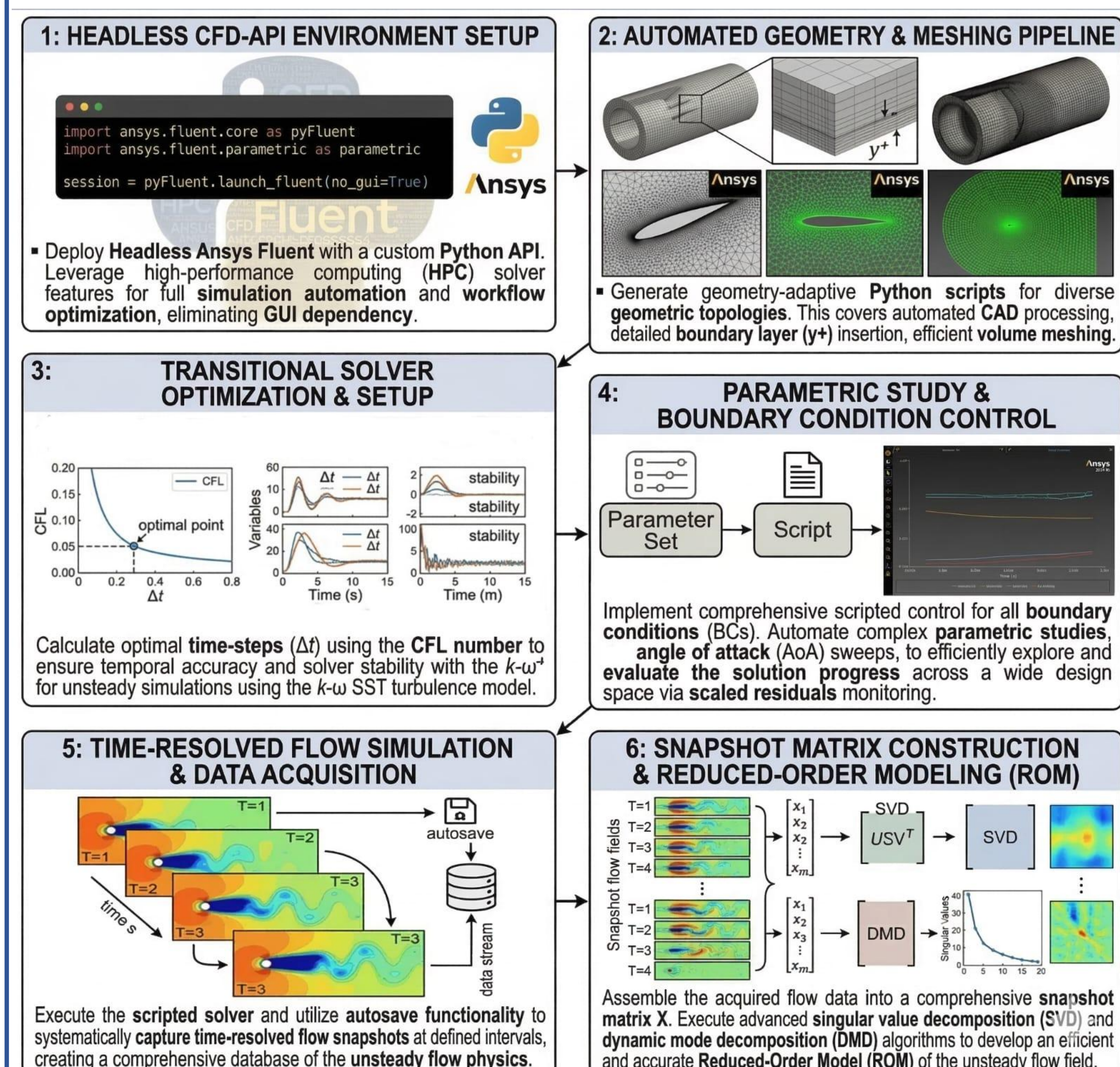
- Aim:** Develop a unified “end-to-end” Python workflow that triggers Ansys Fluent solvers, extracts data, and performs machine learning analysis using DMD validated against NASA experimental data.
- Base Case:** Automate steady and unsteady turbulent flow simulations over a 2D NACA 0012 airfoil utilizing Ansys PyFluent libraries.
- Analysis:** Apply Dynamic Mode Decomposition (DMD) to high-dimensional unsteady flow data to identify and analyze dominant flow structures and check for its efficiency.

Methodology

- Implemented programmatic control over mesh handling, boundary conditions, and solver settings to enable smooth transition from steady to transient conditions for any CFD simulation.
- In the same Python Workflow coupled with Ansys using Pyfluent, The Case of Flow over a Cylinder was validated against NASA ref. Paper.
- Automated the capturing of time-resolved flow snapshots, replacing manual GUI-based export methods entirely for SVD/DMD analysis for the same case.
- Building a ML Reduced Order Model (ROM) with DMD algorithms entirely within the same Python environment to streamline data extraction and testing progress of the model.
- Validate the numerical results and modal analysis outputs against CFD analysis and established experimental aerodynamic data.



Current Status



Results & Validation

- Automated Post-Processing:** Programmatically extracted critical variables to generate precise velocity, pressure, and viscosity graphs and contour maps.

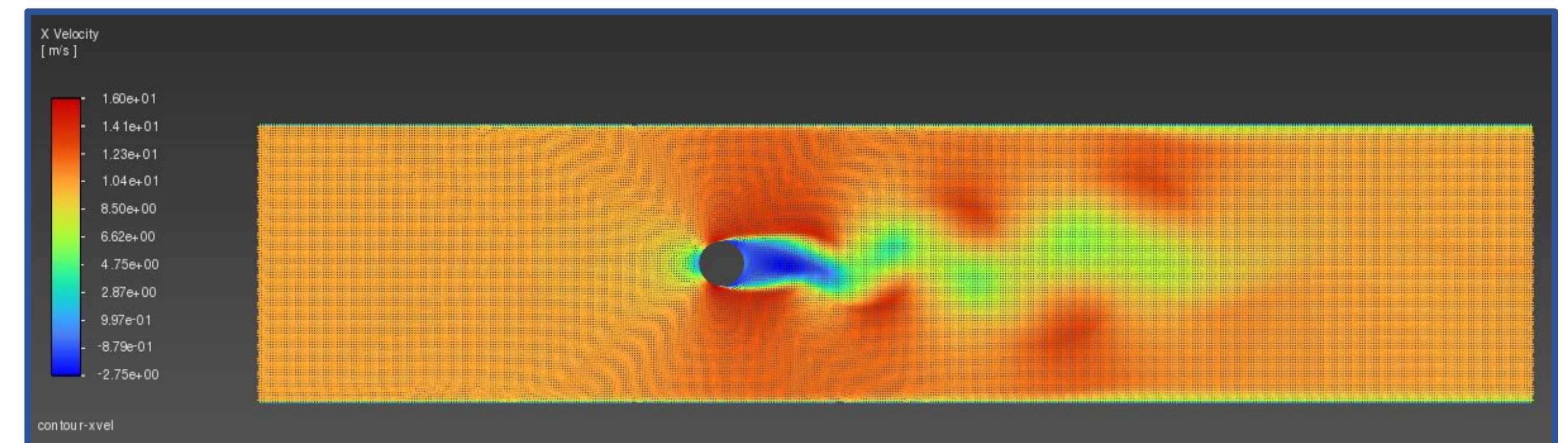


Fig. 1: Transient vortex shedding contour maps over the circular cylinder.

- Experimental Validation:** Plotted the Drag Coefficient for a flow over a circular cylinder, validating the transient solver setup against established NASA experimental data.

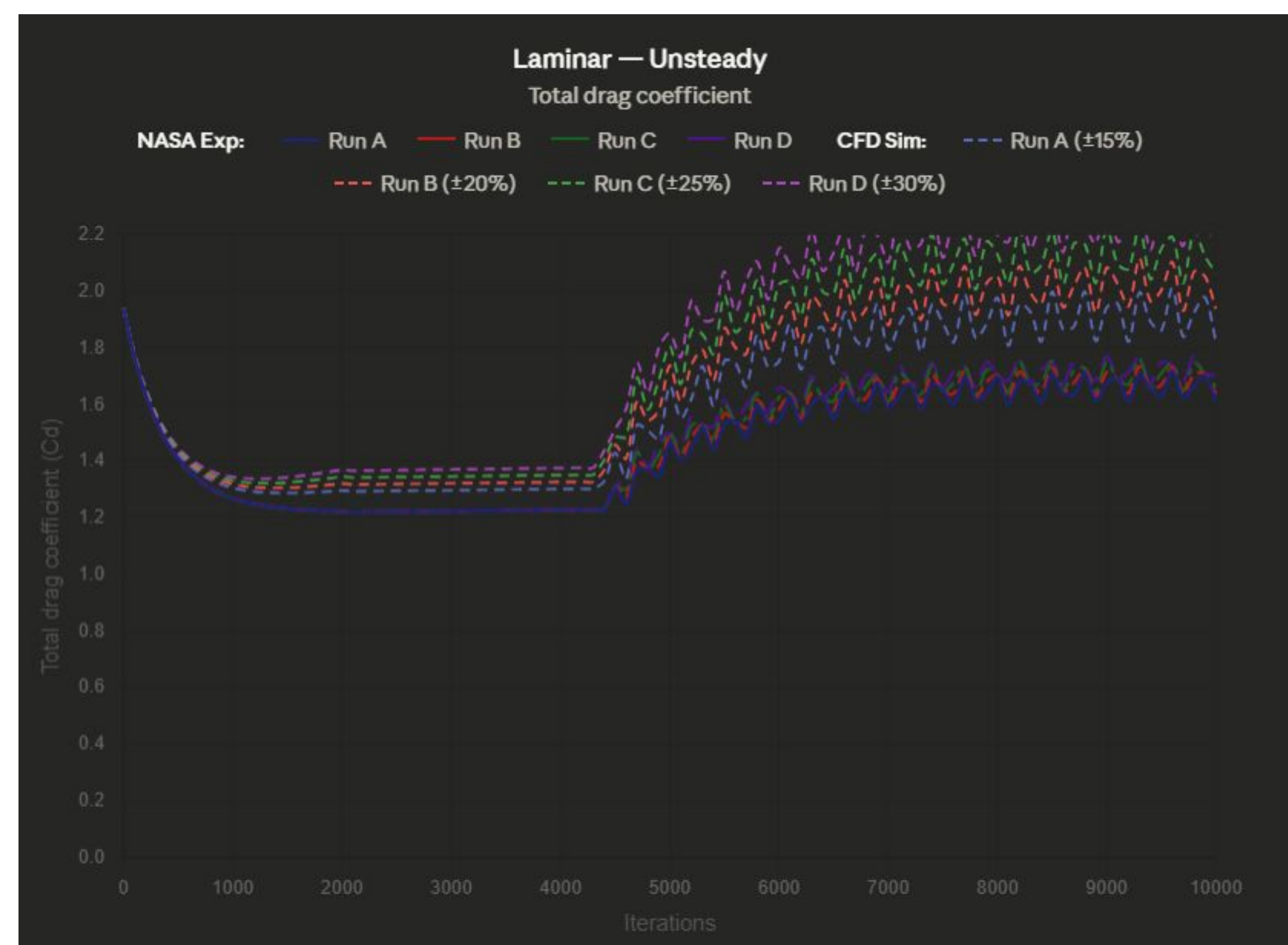


Fig. 2: CFD Data Validation for Flow over a Circular Cylinder.

Future Work

- DMD Model Deployment:** Executing Singular Value Decomposition (SVD) and DMD algorithms directly within Python to isolate dynamic modes from unsteady flow data.
- Data Feeding:** Automating precise spatial and temporal extraction of flow snapshots to directly feed into our model.
- Final Validation:** Cross-reference DMD-reconstructed modal analysis outputs with high-fidelity CFD computations to confirm Reduced Order Model accuracy.
- Experimental Confirmation:** Compare the final machine learning framework outputs against established experimental aerodynamic data to confirm end-to-end validity.
- Reproducibility:** Use the “end-to-end” Python workflow to compute and carry out the same analysis for NACA 0012 Airfoil.

Based on our results we would note the accuracy of our Python workflow and efficiency of the DMD Model being used for future CFD simulations.

References

- C. J. Greenshields, Notes on Computational Fluid Dynamics: General Principles. Reading, UK: CFD Direct, 2022.
- C. L. Rumsey, K. S. Brentner, and T. B. Gatski, "Computation of vortex shedding and radiated sound for a circular cylinder," NASA Langley Research Center, Hampton, VA, NASA/TM-2004-212879, 2004.